

KOCHKI, Russian Federation

WMO: 297240

Lat: 54.336N

Lon: 80.472E

Elev: 528

StdP: 14.42

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-27.7	-39.3	0.6	-33.5	-33.2	0.9	-27.4	28.0	19.3	25.0	16.6	2.4	240	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	85.5	65.4	82.4	64.6	79.5	63.4	68.7	79.4	67.2	77.7	65.7	75.8	8.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	94.7	73.2	63.4	89.1	71.7	61.7	83.8	70.7	33.3	79.2	32.0	77.7	30.9	75.9	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.2	17.4	DB	-36.8	90.6	7.4	3.6	-42.2	93.2	-46.5	95.3	-50.7	97.3	-56.1	100.0	(4)
(5)			WB	-36.8	71.4	7.0	2.1	-41.8	73.0	-45.9	74.2	-49.8	75.4	-54.9	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	-1.1	3.4	19.3	38.8	53.2	63.1	66.2	62.0	50.6	37.7	19.4	5.9		
	DBStd	26.06	15.42	13.56	11.21	10.29	9.61	7.32	5.72	6.67	8.31	9.20	14.08	14.55		
	HDD50	7059	1583	1306	953	360	81	5	0	2	93	391	919	1366		
	HDD65	11151	2048	1726	1418	786	379	119	54	139	437	845	1369	1831		
	CDD50	1598	0	0	0	23	180	397	503	373	111	11	0	0		
	CDD65	216	0	0	0	0	13	62	92	44	5	0	0	0		
	CDH74	2264	0	0	0	9	254	624	778	499	100	0	0	0		
	CDH80	547	0	0	0	1	61	147	185	126	27	0	0	0		
	WSAvg	7.1	7.7	7.6	7.8	7.6	7.5	6.2	5.4	5.6	6.2	7.1	8.4	8.6		
	PrecAvg	15.2	0.7	0.6	0.7	0.8	1.3	1.9	2.7	1.8	1.4	1.4	1.2	1.0		
	PrecMax	20.9	1.3	0.9	1.5	1.9	2.9	3.3	6.1	3.9	2.5	3.2	2.4	1.9		
(15) Precipitation	PrecMin	10.8	0.1	0.0	0.1	0.1	0.1	0.4	0.7	0.5	0.2	0.5	0.7	0.4		
	PrecStd	2.8	0.4	0.2	0.4	0.4	0.7	0.7	1.4	0.9	0.7	0.7	0.4	0.4		
	DB	33.5	33.5	44.3	74.2	86.8	87.8	89.3	88.5	84.0	67.9	47.4	34.1			
	MCWB	32.0	31.5	37.9	54.4	61.7	66.3	68.7	66.8	61.0	52.7	42.9	32.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	28.6	30.3	39.5	66.8	80.4	84.2	84.7	83.7	75.6	61.6	42.5	32.0			
	2% MCWB	27.3	28.5	35.1	51.7	58.9	64.3	67.6	66.1	58.2	49.5	39.2	30.9			
	5% DB	25.0	27.0	36.8	61.1	76.0	81.1	81.5	79.6	70.5	55.9	38.7	28.8			
	5% MCWB	23.6	25.5	33.5	49.0	57.0	63.7	66.3	64.0	56.4	46.9	36.2	27.4			
	10% DB	21.4	23.5	34.3	55.4	70.9	77.5	78.3	75.8	65.5	50.9	35.3	25.3			
	10% MCWB	20.2	22.1	31.8	45.8	54.8	62.6	65.3	62.0	53.6	44.3	33.2	23.9			
	0.4% WB	32.4	31.5	38.5	56.4	64.0	68.7	71.8	70.2	63.1	53.8	43.7	32.7			
	0.4% MCDB	33.6	32.7	43.0	70.2	82.3	80.7	83.1	81.8	76.3	64.6	47.0	33.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	27.5	28.7	35.7	53.0	60.9	66.8	69.5	67.5	59.6	50.5	39.5	30.9			
	2% MCDB	28.7	30.1	38.8	64.8	76.7	79.1	79.8	79.2	72.4	59.2	42.0	31.9			
	5% WB	23.8	25.7	33.8	49.9	58.3	65.3	68.0	65.6	57.1	47.7	36.1	27.6			
	5% MCDB	25.0	27.0	36.4	59.5	72.3	77.1	77.9	76.3	69.0	54.7	38.4	28.8			
	10% WB	20.1	22.2	32.0	46.5	56.1	63.8	66.7	63.6	54.5	44.9	33.3	24.0			
	10% MCDB	21.3	23.5	34.4	55.0	69.2	74.7	75.9	73.7	63.9	50.7	35.2	25.3			
	MDBR	14.9	16.5	16.7	17.7	21.7	19.2	18.4	20.3	19.5	15.2	12.0	13.9			
	5% DB	15.1	15.4	15.6	27.3	29.2	25.3	23.2	26.5	28.7	24.9	11.5	13.6			
(36) Mean Daily Temperature Range	5% WB	15.1	14.8	12.4	15.1	12.8	10.3	9.1	11.2	13.6	14.6	9.4	13.5			
	MCDBR	14.9	15.0	14.0	25.3	25.3	21.3	19.4	22.0	25.5	22.3	11.2	13.3			
	MCWBR	14.9	14.4	11.6	15.0	12.4	10.0	9.1	10.4	12.8	14.2	9.6	13.4			
	MCWBR	14.9	14.4	11.6	15.0	12.4	10.0	9.1	10.4	12.8	14.2	9.6	13.4			
(40) Clear-Sky Solar Irradiance	taub	0.292	0.324	0.377	0.405	0.407	0.408	0.434	0.414	0.368	0.351	0.327	0.295			
	taud	2.021	1.974	1.911	2.117	2.202	2.280	2.202	2.235	2.315	2.235	2.068	1.971			
	Ebn at Noon	194	230	244	258	266	267	256	253	251	223	177	159			
	Edh at Noon	23	34	46	43	42	40	42	38	31	27	22	20			
(44) All-Sky Solar Radiation	RadAvg	323	657	1122	1499	1767	1933	1739	1484	1027	525	255	212			
	RadStd	26	37	79	94	184	166	113	100	110	69	42	30			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page